

Project Proposal: SmartHydro

Distributed System for Automated Hydroponic Greenhouses

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1. Problem Statement and Scope

Managing multiple hydroponic crops simultaneously requires precise, time-varying environmental control that traditional monolithic systems struggle to provide, particularly when handling hardware faults. To address this, the “SmartHydro” project proposes a robust Edge-to-Cloud software system strictly focused on automating and dynamically managing different crops in parallel, with a highly scalable architecture designed for future expansion to multiple independent greenhouses.

The system is divided into two main components: a **Cloud Backend** that allows the Agronomist to define structured “Cultivation Recipes” (target setpoints for temperature, humidity, pH) via a Web Dashboard, and an **Edge Controller** operating directly in the greenhouse. The Edge Controller executes real-time control loops by acquiring sensor data and driving actuators to track the recipes. The project scope encompasses both nominal concurrent multi-variable control and the active simulation and automatic recovery of hardware failures, ensuring crop safety without constant human supervision.

2. Programming Language

The Edge Controller will be developed in **C++** to ensure high performance and reliability for the real-time simulated control loops. The Cloud Backend will be implemented in **Python** to manage business logic and data routing, while the Agronomist Web Dashboard will utilize **HTML, CSS, and JavaScript** for an interactive and accessible user interface.

3. Core Features & System Dynamics

To ensure a robust and flexible simulation of a real-world automated greenhouse, the system will implement the following core features:

- **Multi-Crop Management:** Simultaneous management of multiple independent greenhouse zones, executing different cultivation recipes in parallel.
- **Dynamic Control Strategies:** The ability for the Agronomist to switch environmental control policies (e.g., simplified PID, threshold-based control, predictive control) at runtime.
- **Fault Simulation & Recovery:** Intentional simulation of hardware malfunctions (e.g., sensor failure, actuator breakdown) to trigger automatic safety procedures, degraded modes, and emergency lockdowns.
- **Telemetry & Event Notifications:** Continuous logging of environmental data and advanced dispatching of real-time alerts and state-change events to the central dashboard.

4. Architectural & Design Choices

The software architecture will rely explicitly on specific Design Patterns to guarantee modularity, maintainability, and a clear separation of concerns:

- **State:** Employed to manage the Finite State Machines (FSMs) dictating the crop’s lifecycle phases (Germination, Vegetative, Flowering/Fruiting, Harvest) and the system’s operational states (Nominal, Degraded, Emergency Lockdown).
- **Strategy:** Utilized to encapsulate the different environmental control algorithms, making them safely interchangeable at runtime.
- **Observer:** Implemented to handle the reactive telemetry system, propagating sensor updates, alarms, and state changes from the Edge Controller to the Dashboard.
- **Adapter:** Used to decouple the core control logic from the physical hardware, facilitating the current use of software simulators and ensuring future drop-in compatibility with real physical sensors and actuators.